

‘Home work problems in Data structure’

1. Write a C program to store n elements in an array using dynamic memory allocation and find the largest number.

```
main.c Run ▶ Output
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 int main()
5 {
6     int n, i, largest;
7     int *a;
8
9     printf("Enter number of elements: ");
10    scanf("%d", &n);
11
12    a = (int *)malloc(n * sizeof(int));
13
14    printf("Enter elements:\n");
15    for(i = 0; i < n; i++)
16        scanf("%d", &a[i]);
17
18    largest = a[0];
19
20    for(i = 1; i < n; i++)
21    {
22        if(a[i] > largest)
23            largest = a[i];
24    }
25
26    printf("Largest number = %d", largest);
27
28    free(a);
29
30    return 0;
31 }
```

Enter number of elements: 4
Enter elements:
2
3
5
8
Largest number = 8

=== Code Execution Successful ===

2. Print an array in reverse order using pointers.

```
main.c Run ▶ Output
1 #include <stdio.h>
2
3 int main()
4 {
5     int a[5], i;
6     int *p;
7
8     p = a;
9
10    printf("Enter 5 elements:\n");
11    for(i = 0; i < 5; i++)
12        scanf("%d", p + i);
13
14    printf("Array in reverse order:\n");
15
16    for(i = 4; i >= 0; i--)
17        printf("%d ", *(p + i));
18
19    return 0;
20 }
```

Enter 5 elements:
3
4
7
8
9
Array in reverse order:
9 8 7 4 3

=== Code Execution Successful ===

3. Write a C program to store information of students using structure. Sort them by name.

```
main.c Run ▶ Output
1 #include <stdio.h>
2 #include <string.h>
3
4 struct Student
5 {
6     char name[50];
7     int roll;
8     float marks;
9 };
10
11 int main()
12 {
13     struct Student s[50], temp;
14     int n, i, j;
15
16     printf("Enter number of students: ");
17     scanf("%d", &n);
18
19     for(i = 0; i < n; i++)
20     {
21         printf("\nEnter details of student %d\n", i + 1);
22
23         printf("Name: ");
24         scanf("%s", s[i].name);
25
26         printf("Roll number: ");
27         scanf("%d", &s[i].roll);
28
29         printf("Marks: ");
30         scanf("%f", &s[i].marks);
31     }
32
33     for(i = 0; i < n - 1; i++)
34     {
35         for(j = i + 1; j < n; j++)
36         {
37             if(strcmp(s[i].name, s[j].name) > 0)
38             {
39                 temp = s[i];
40                 s[i] = s[j];
41                 s[j] = temp;
42             }
43         }
44     }
45
46     printf("\nStudents sorted by name:\n");
47     for(i = 0; i < n; i++)
48     {
49         printf("Name: %s", s[i].name);
50         printf("\nRoll Number: %d", s[i].roll);
51         printf("\nMarks: %.2f\n", s[i].marks);
52     }
53
54     return 0;
55 }
```

```
Enter number of students: 2
Enter details of student 1
Name: Farah
Roll number: 90
Marks: 98
Enter details of student 2
Name: Basith
Roll number: 14
Marks: 99
Students sorted by name:
Name: Basith
Roll Number: 14
Marks: 99.00
Name: Farah
Roll Number: 90
Marks: 98.00
--- Code Execution Successful ---
```

4. Write a C program to add n distances given in inch-feet system using structure.

```
main.c Run ▶ Output
1 #include <stdio.h>
2
3 struct Distance
4 {
5     int feet;
6     float inch;
7 };
8
9 int main()
10 {
11     struct Distance d[50], sum = {0, 0};
12     int n, i;
13
14     printf("Enter number of distances: ");
15     scanf("%d", &n);
16
17     for(i = 0; i < n; i++)
18     {
19         printf("\nEnter distance %d\n", i + 1);
20
21         printf("Feet: ");
22         scanf("%d", &d[i].feet);
23
24         printf("Inches: ");
25         scanf("%f", &d[i].inch);
26
27         sum.feet += d[i].feet;
28         sum.inch += d[i].inch;
29     }
30
31     /* Convert inches into feet */
32     while(sum.inch >= 12)
33     {
34         sum.inch -= 12;
35         sum.feet++;
36     }
37
38     printf("\nTotal distance = %d feet %.2f inches",
39         sum.feet, sum.inch);
40
41     return 0;
42 }
```

```
Enter number of distances: 1
Enter distance 1
Feet: 5
Inches: 8
Total distance = 5 feet 8.00 inches
--- Code Execution Successful ---
```

5. Write a C program to add two complex numbers by passing structure to a function.

```

main.c Run ▶ Output
1 #include <stdio.h>
2
3 struct Complex
4 {
5     float real;
6     float imag;
7 };
8
9 struct Complex add(struct Complex c1, struct Complex c2)
10 {
11     struct Complex result;
12
13     result.real = c1.real + c2.real;
14     result.imag = c1.imag + c2.imag;
15
16     return result;
17 }
18
19 int main()
20 {
21     struct Complex c1, c2, sum;
22
23     printf("Enter first complex number:\n");
24     printf("Real part: ");
25     scanf("%f", &c1.real);
26
27     printf("Imaginary part: ");
28     scanf("%f", &c1.imag);
29
30     printf("\nEnter second complex number:\n");
31     printf("Real part: ");
32     scanf("%f", &c2.real);
33
34     printf("Imaginary part: ");
35     scanf("%f", &c2.imag);
36
37     sum = add(c1, c2);
38
39     printf("\nSum = %.2f + %.2fi", sum.real, sum.imag);
40
41     return 0;
42 }

```

Output

```

Enter first complex number:
Real part: 3
Imaginary part: 4

Enter second complex number:
Real part: 5
Imaginary part: 2

Sum = 8.00 + 6.00i

=== Code Execution Successful ===

```

6. Write a C program to maintain inventory management system using structure.

```

main.c Run ▶ Output
1 #include <stdio.h>
2
3 struct Item
4 {
5     int id;
6     char name[50];
7     int quantity;
8     float price;
9 };
10
11 int main()
12 {
13     struct Item item[50];
14     int n = 0, choice, i, id;
15
16     do
17     {
18         printf("\n\n--- INVENTORY MANAGEMENT ---");
19         printf("\n1. Add Item");
20         printf("\n2. Display Items");
21         printf("\n3. Search Item");
22         printf("\n4. Exit");
23
24         printf("\nEnter your choice: ");
25         scanf("%d", &choice);
26
27         switch(choice)
28         {
29             case 1:
30                 printf("\nEnter Item ID: ");
31                 scanf("%d", &item[n].id);
32
33                 printf("\nEnter Item Name: ");
34                 scanf("%s", item[n].name);
35
36                 printf("\nEnter Quantity: ");
37                 scanf("%d", &item[n].quantity);
38
39                 printf("\nEnter Price: ");
40                 scanf("%f", &item[n].price);
41
42                 n++;

```

Output

```

--- INVENTORY MANAGEMENT ---
1. Add Item
2. Display Items
3. Search Item
4. Exit
Enter your choice: 1

Enter Item ID: 101
Enter Item Name: pen
Enter Quantity: 50
Enter Price: 10
Item added successfully.

--- INVENTORY MANAGEMENT ---
1. Add Item
2. Display Items
3. Search Item
4. Exit
Enter your choice: 2

--- INVENTORY ---
ID: 101
Name: pen
Quantity: 50
Price: 10.00

--- INVENTORY MANAGEMENT ---
1. Add Item
2. Display Items
3. Search Item
4. Exit
Enter your choice: 4

Exiting...

=== Code Execution Successful ===

```

```

main.c Run Output
12
13 printf("Enter Item Name: ");
14 scanf("%s", item[0].name);
15
16 printf("Enter Quantity: ");
17 scanf("%d", &item[0].quantity);
18
19 printf("Enter Price: ");
20 scanf("%f", &item[0].price);
21
22 //
23 printf("Item added successfully.");
24 break;
25
26 case 2:
27 printf("\n--- INVENTORY ---\n");
28 for(i = 0; i < n; i++)
29 {
30 printf("\nID: %d", item[i].id);
31 printf("\nName: %s", item[i].name);
32 printf("\nQuantity: %d", item[i].quantity);
33 printf("\nPrice: %.2f", item[i].price);
34 }
35 break;
36
37 case 3:
38 printf("Enter Item ID to search: ");
39 scanf("%d", &id);
40
41 for(i = 0; i < n; i++)
42 {
43 if(item[i].id == id)
44 {
45 printf("Item Found");
46 printf("\nName: %s", item[i].name);
47 printf("\nQuantity: %d", item[i].quantity);
48 printf("\nPrice: %.2f", item[i].price);
49 break;
50 }
51 }
52
53 if(i == n)
54 printf("Item not found.");
55 break;
56
57 case 4:
58 printf("Exiting...");
59 break;
60
61 default:
62 printf("Invalid choice.");
63 }
64 } while(choice != 4);
65
66 return 0;
67 }

```

```

--- INVENTORY MANAGEMENT ---
1. Add Item
2. Display Items
3. Search Item
4. Exit
Enter your choice: 1
Enter Item ID: 101
Enter Item Name: pen
Enter Quantity: 50
Enter Price: 10
Item added successfully.
--- INVENTORY MANAGEMENT ---
1. Add Item
2. Display Items
3. Search Item
4. Exit
Enter your choice: 2
--- INVENTORY ---
ID: 101
Name: pen
Quantity: 50
Price: 10.00
--- INVENTORY MANAGEMENT ---
1. Add Item
2. Display Items
3. Search Item
4. Exit
Enter your choice: 4
Exiting...
=== Code Execution Successful ===

```

7. Write a C program for decimal to binary, octal and hexadecimal conversion.

```

main.c Run Output
1 #include <stdio.h>
2
3 void binary(int n)
4 {
5     int a[32], i = 0;
6
7     if(n == 0)
8     {
9         printf("0");
10        return;
11    }
12
13    while(n > 0)
14    {
15        a[i] = n % 2;
16        n = n / 2;
17        i++;
18    }
19
20    while(i > 0)
21    {
22        i--;
23        printf("%d", a[i]);
24    }
25 }
26
27 int main()
28 {
29     int n;
30
31     printf("Enter a decimal number: ");
32     scanf("%d", &n);
33
34     printf("Binary      = ");
35     binary(n);
36
37     printf("\nOctal      = %o", n);
38     printf("\nHexadecimal = %X", n);
39
40     return 0;
41 }

```

```

Enter a decimal number: 25.5
Binary      = 11001
Octal      = 31
Hexadecimal = 19
=== Code Execution Successful ===

```

8. Write a C program to calculate difference between two time periods.

```
main.c Run ▶ Output
1 #include <stdio.h>
2
3 struct Time
4 {
5     int hour;
6     int minute;
7     int second;
8 };
9
10 int main()
11 {
12     struct Time t1, t2, diff;
13
14     printf("Enter first time (HH MM SS): ");
15     scanf("%d %d %d", &t1.hour, &t1.minute, &t1.second);
16
17     printf("Enter second time (HH MM SS): ");
18     scanf("%d %d %d", &t2.hour, &t2.minute, &t2.second);
19
20     if(t2.second < t1.second)
21     {
22         t2.second += 60;
23         t2.minute--;
24     }
25
26     diff.second = t2.second - t1.second;
27
28     if(t2.minute < t1.minute)
29     {
30         t2.minute += 60;
31         t2.hour--;
32     }
33
34     diff.minute = t2.minute - t1.minute;
35     diff.hour = t2.hour - t1.hour;
36
37     printf("\nTime difference = %02d:%02d:%02d",
38         diff.hour, diff.minute, diff.second);
39
40     return 0;
41 }
```

Enter first time (HH MM SS): 10 30 45
Enter second time (HH MM SS): 12 45 20
Time difference = 02:14:35
=== Code Execution Successful ===

9. Write a C program to implement an employee management system using structures.

```
main.c Run ▶ Output
1 #include <stdio.h>
2
3 struct Employee
4 {
5     int id;
6     char name[50];
7     float salary;
8 };
9
10 int main()
11 {
12     struct Employee e[50];
13     int n, i, searchid, found = 0;
14
15     printf("Enter number of employees: ");
16     scanf("%d", &n);
17
18     for(i = 0; i < n; i++)
19     {
20         printf("\nEnter details of employee %d\n", i + 1);
21
22         printf("ID: ");
23         scanf("%d", &e[i].id);
24
25         printf("Name: ");
26         scanf("%s", e[i].name);
27
28         printf("Salary: ");
29         scanf("%f", &e[i].salary);
30     }
31
32     printf("\n--- EMPLOYEE DETAILS ---\n");
33
34     for(i = 0; i < n; i++)
35     {
36         printf("\nID: %d", e[i].id);
37         printf("\nName: %s", e[i].name);
38         printf("\nSalary: %.2f\n", e[i].salary);
39     }
40
41     printf("\nEnter employee ID to search: ");
42     scanf("%d", &searchid);
```

Enter number of employees: 2
Enter details of employee 1
ID: 101
Name: farah
Salary: 100000
Enter details of employee 2
ID: 102
Name: basith
Salary: 500000
--- EMPLOYEE DETAILS ---
ID: 101
Name: farah
Salary: 100000.00
ID: 102
Name: basith
Salary: 500000.00
Enter employee ID to search: |

```

main.c Run
20 printf("Enter details of employee again , i + 1);
21
22 printf("ID: ");
23 scanf("%d", &e[i].id);
24
25 printf("Name: ");
26 scanf("%s", e[i].name);
27
28 printf("Salary: ");
29 scanf("%f", &e[i].salary);
30 }
31
32 printf("\n--- EMPLOYEE DETAILS ---\n");
33
34 for(i = 0; i < n; i++)
35 {
36     printf("\nID: %d", e[i].id);
37     printf("\nName: %s", e[i].name);
38     printf("\nSalary: %.2f\n", e[i].salary);
39 }
40
41 printf("\nEnter employee ID to search: ");
42 scanf("%d", &searchId);
43
44 for(i = 0; i < n; i++)
45 {
46     if(e[i].id == searchId)
47     {
48         printf("\nEmployee Found!");
49         printf("\nID: %d", e[i].id);
50         printf("\nName: %s", e[i].name);
51         printf("\nSalary: %.2f", e[i].salary);
52
53         found = 1;
54         break;
55     }
56 }
57
58 if(found == 0)
59     printf("\nEmployee not found.");
60
61 return 0;
62 }

```

Output

```

Enter number of employees: 2

Enter details of employee 1
ID: 101
Name: farah
Salary: 100000

Enter details of employee 2
ID: 102
Name: basith
Salary: 500000

--- EMPLOYEE DETAILS ---

ID: 101
Name: farah
Salary: 100000.00

ID: 102
Name: basith
Salary: 500000.00

Enter employee ID to search: |

```

10. Write a C program to calculate area and perimeter of a rectangle and triangle using function pointers.

```

main.c Run
1 #include <stdio.h>
2 #include <math.h>
3
4 float rectangleArea(float l, float b)
5 {
6     return l * b;
7 }
8
9 float rectanglePerimeter(float l, float b)
10 {
11     return 2 * (l + b);
12 }
13
14 float triangleArea(float b, float h)
15 {
16     return 0.5 * b * h;
17 }
18
19 float trianglePerimeter(float a, float b, float c)
20 {
21     return a + b + c;
22 }
23
24 int main()
25 {
26     int choice;
27
28     float (*area)(float, float);
29     float (*perimeter)(float, float);
30
31     float l, b, h, a, c;
32
33     printf("1. Rectangle");
34     printf("\n2. Triangle");
35
36     printf("\nEnter your choice: ");
37     scanf("%d", &choice);
38
39     if(choice == 1)
40     {
41         printf("\nEnter length and breadth: ");
42         scanf("%f %f", &l, &b);
43

```

Output

```

1. Rectangle
2. Triangle
Enter your choice: 1

Enter length and breadth: 2 4

Area = 8.00
Perimeter = 12.00

=== Code Execution Successful ===

```

```
main.c Run Output
28 float (*area)(float, float);
29 float (*perimeter)(float, float);
30
31 float l, b, h, a, c;
32
33 printf("1. Rectangle");
34 printf("\n2. Triangle");
35
36 printf("\nEnter your choice: ");
37 scanf("%d", &choice);
38
39 if(choice == 1)
40 {
41     printf("\nEnter length and breadth: ");
42     scanf("%f %f", &l, &b);
43
44     area = rectangleArea;
45     perimeter = rectanglePerimeter;
46
47     printf("\nArea = %.2f", area(l, b));
48     printf("\nPerimeter = %.2f", perimeter(l, b));
49 }
50 else if(choice == 2)
51 {
52     printf("\nEnter base and height: ");
53     scanf("%f %f", &b, &h);
54
55     printf("Enter other two sides: ");
56     scanf("%f %f", &a, &c);
57
58     area = triangleArea;
59
60     printf("\nArea = %.2f", area(b, h));
61     printf("\nPerimeter = %.2f", a + b + c);
62 }
63 else
64 {
65     printf("Invalid choice.");
66 }
67
68 return 0;
69 }
```

1. Rectangle
2. Triangle
Enter your choice: 1

Enter length and breadth: 2 4

Area = 8.00
Perimeter = 12.00

=== Code Execution Successful ===